

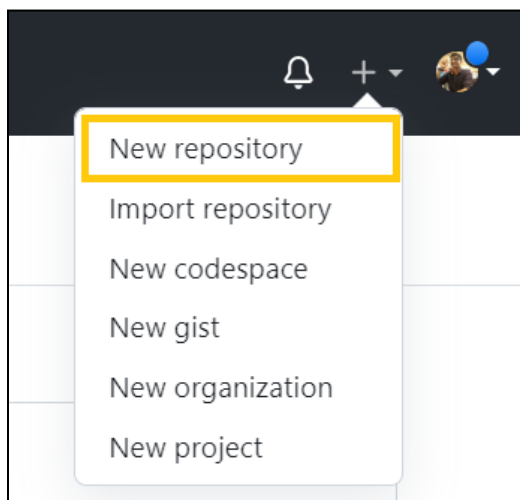
Stock Prices Insights Generation Application

Procedure in a nutshell

1. Download the Application folder shared along with this document and unzip it
2. Create a new GitHub repository
3. Upload the content of Application folder to this repository
4. Add OpenAI API key into Codespaces Secrets of this repository
5. Create a new Codespace
6. Create and activate virtual environment
7. Install requirements
8. Run main.py file to load data from pickle files and save it into a SQLite DB
9. Run app.py to launch the application
10. Once done, close the application and delete the codespace

Steps

1. First, create an account on [GitHub](https://github.com) if you don't have one already.
2. Once the account is created, create a new repository



3. Fill in details for your repository, and click Create. Give only the repository name, as the downloaded folder already has the README and .gitignore files.



Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

Required fields are marked with an asterisk (*).

Repository template

No template ▾

Start your repository with a template repository's contents.

Owner *

 yrajm1997 ▾

Repository name *

/ stock-project

✔ stock-project is available.

Great repository names are short and memorable. Need inspiration? How about **psychic-computing-machine** ?

Description (optional)



Public

Anyone on the internet can see this repository. You choose who can commit.



Private

You choose who can see and commit to this repository.

Initialize this repository with:

☐ **Add a README file**

This is where you can write a long description for your project. [Learn more about READMEs.](#)

Add .gitignore

.gitignore template: None ▾

Choose which files not to track from a list of templates. [Learn more about ignoring files.](#)

Choose a license

License: None ▾

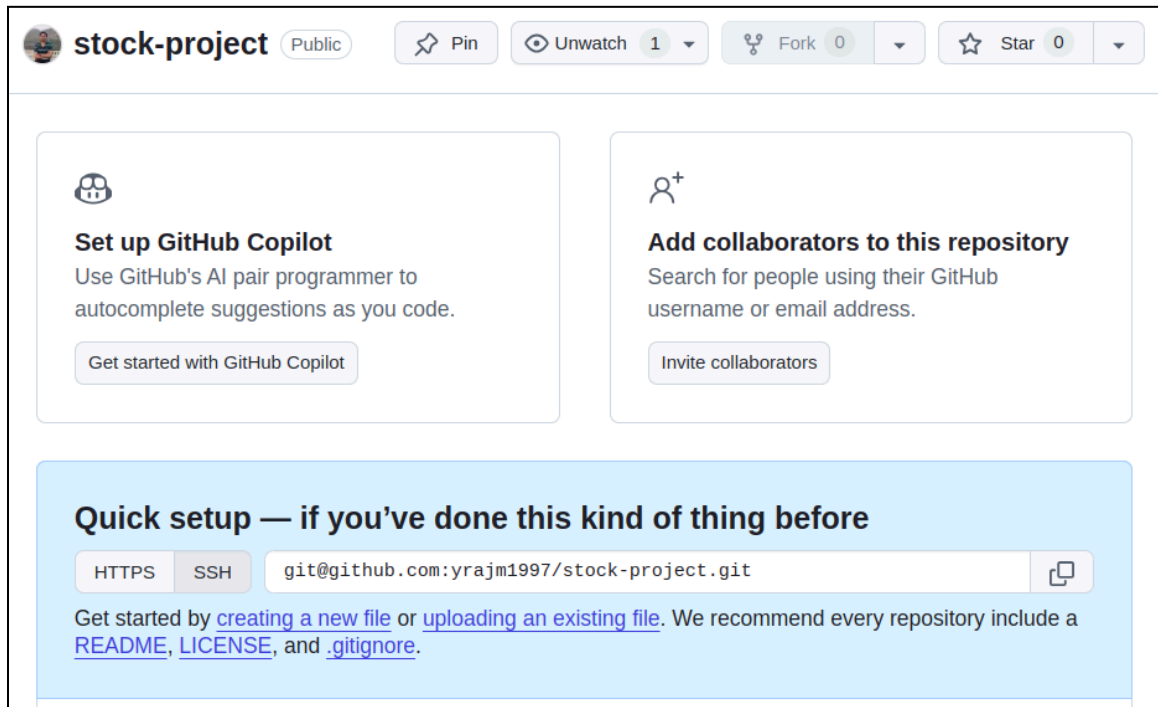
A license tells others what they can and can't do with your code. [Learn more about licenses.](#)

 You are creating a public repository in your personal account.

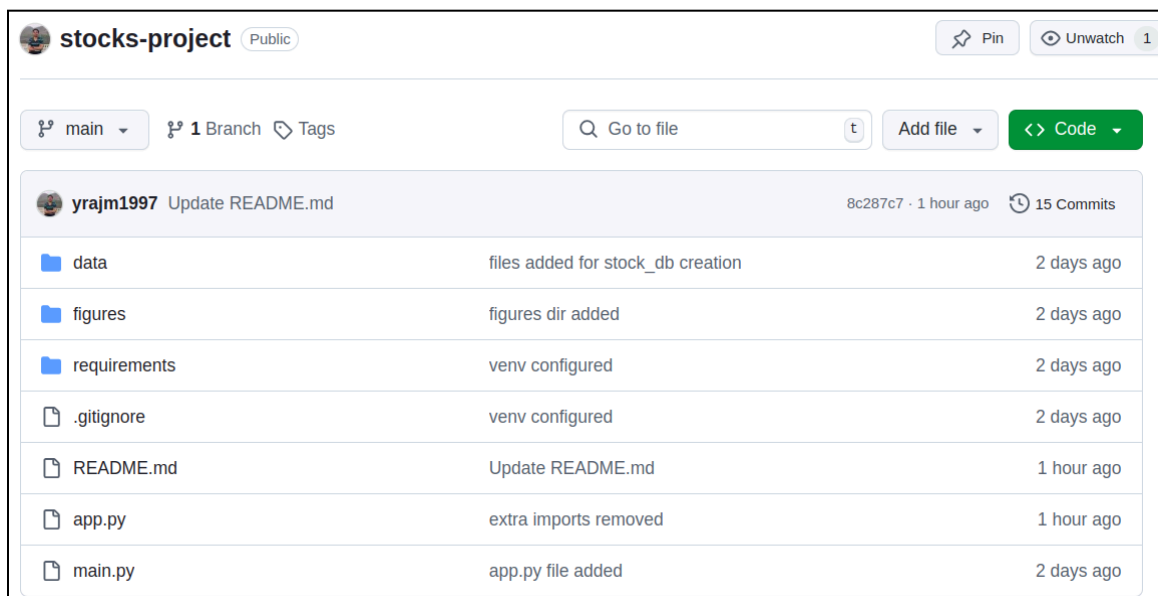
Create repository



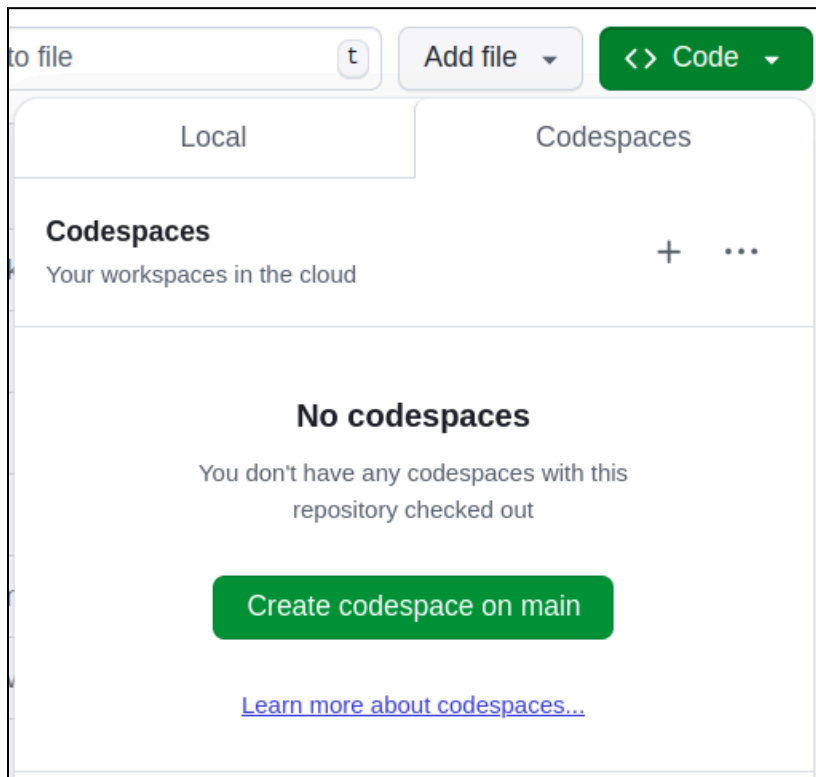
- Once the repository is created, clone it into your system using the SSH link showing in your repo. For example:



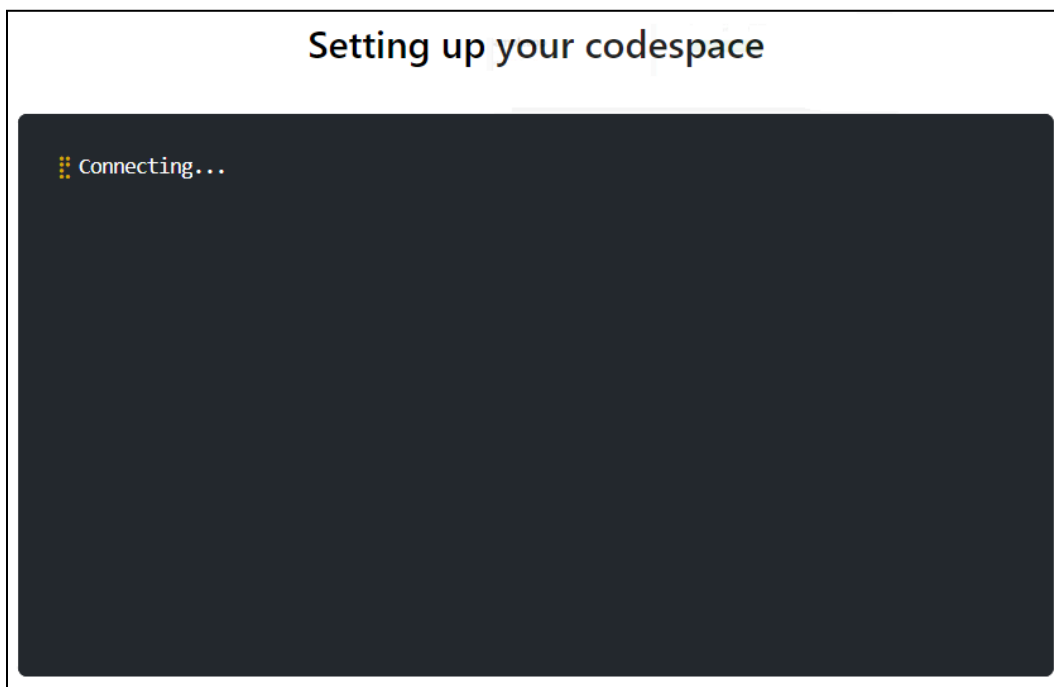
- Download the Application folder shared along with this document and unzip it. After cloning your GitHub repository, copy and paste the files from the Application folder to your cloned repo, then commit and push the changes.
- Application files should be visible on the remote repository in GitHub.



- Now, to start a new Codespace, go to the 'Code' dropdown. Select the Codespaces tab, and click on 'Create codespace on main'.

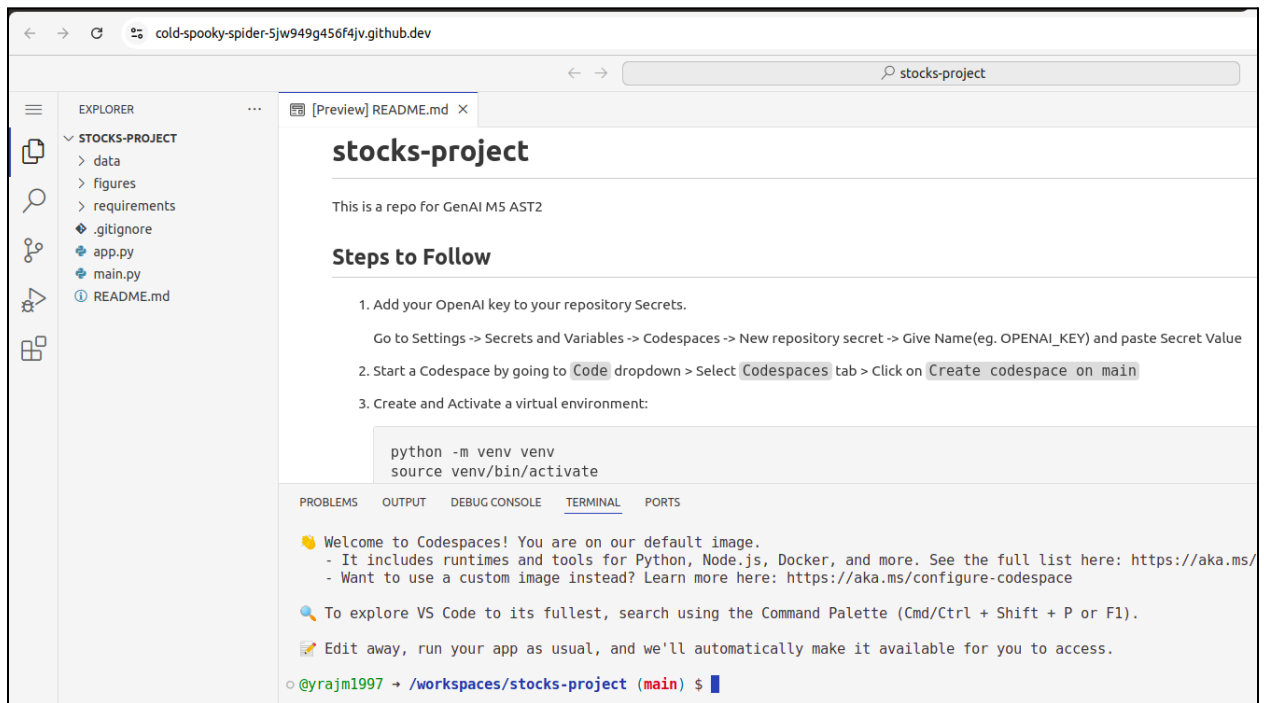


- It will open a new tab, and start *Setting up your codespace*.

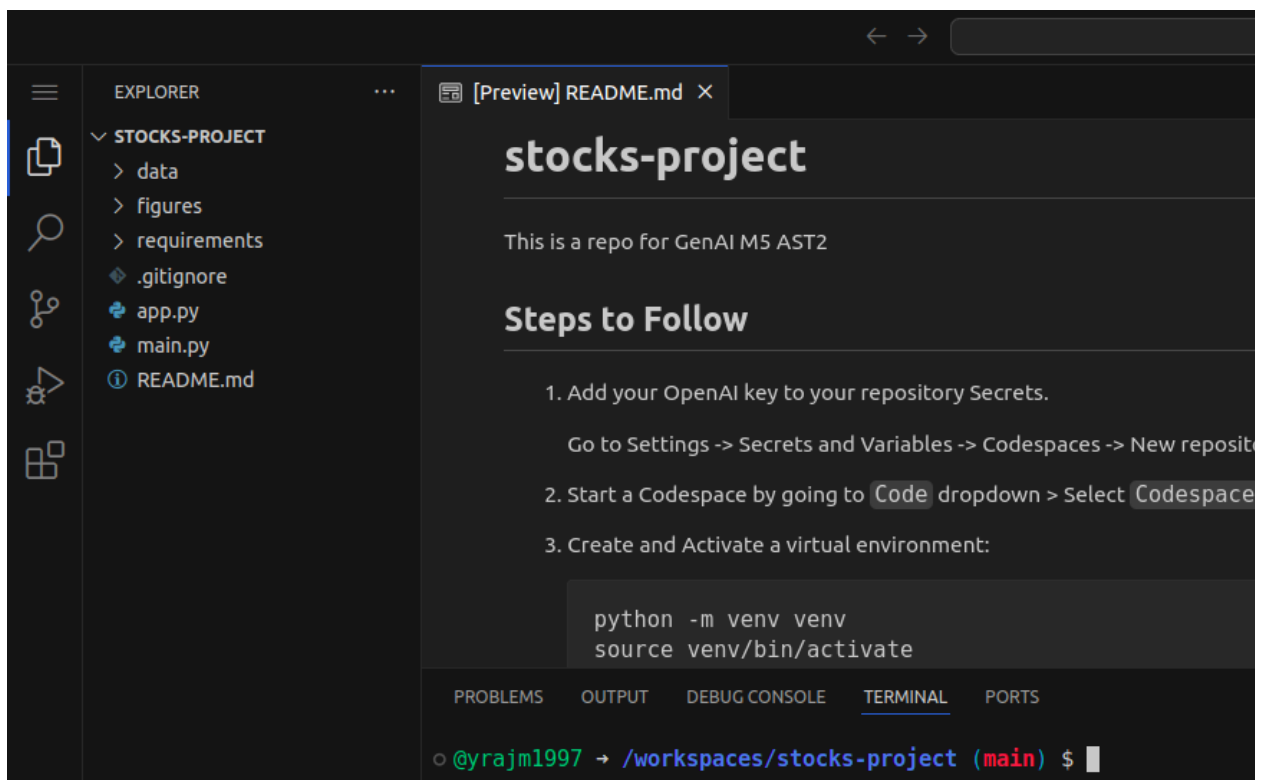




9. After loading, the Editor should open in the browser similar to below.



10. By default it is Light mode. You can change the theme by going to Files -> Preferences -> Theme -> Color theme -> Default Dark Modern.





11. Next, create and activate a virtual environment by running the below commands:

```
python -m venv venv  
source venv/bin/activate
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
  
● @yrajm1997 → /workspaces/stocks-project (main) $ python -m venv venv  
○ @yrajm1997 → /workspaces/stocks-project (main) $  
● @yrajm1997 → /workspaces/stocks-project (main) $ source venv/bin/activate  
○ (venv) @yrajm1997 → /workspaces/stocks-project (main) $  
○ (venv) @yrajm1997 → /workspaces/stocks-project (main) $
```

12. Install requirements (*This step will take time*)

```
pip install -r requirements/requirements.txt
```

```
○ (venv) @yrajm1997 → /workspaces/stocks-project (main) $ pip install -r requirements/requirements.txt  
Collecting numpy (from -r requirements/requirements.txt (line 1))  
  Obtaining dependency information for numpy from https://files.pythonhosted.org/packages/9b/b4/e3c7e6  
p312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata  
  Downloading numpy-2.1.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (60 kB)  
    60.9/60.9 kB 723.8 kB/s eta 0:00:00  
Collecting pandas (from -r requirements/requirements.txt (line 2))
```

13. Run the main.py file to load the data from pickle files and save it inside a SQLite DB.

```
python main.py
```

```
● (venv) @yrajm1997 → /workspaces/stocks-project (main) $ python main.py  
File 'stock_db.sqlite' does not exist. Creating new...  
Created DB successfully!  
  
Table 'stock_prices' created successfully!  
Data inserted into 'stock_prices' successfully!  
  
Table 'income_statement' created successfully  
Data inserted into 'income_statement' successfully!  
  
Table 'balancesheet_statement' created successfully  
Data inserted into 'balancesheet_statement' successfully!  
  
Table 'cashflow_statement' created successfully  
Data inserted into 'cashflow_statement' successfully!  
DB connection closed!
```

After this step, you should have the 'stock_db.sqlite' file available within the codespace.



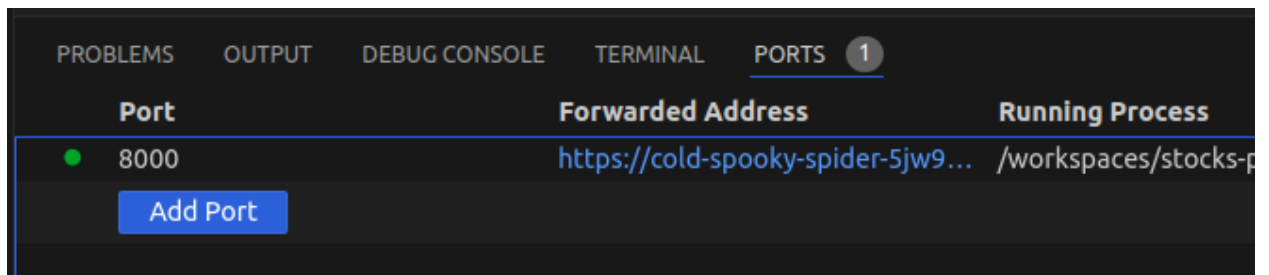
14. To start the application run the following command:

```
chainlit run app.py
```

```
o (venv) @yrajm1997 → /workspaces/stocks-project (main) $ chainlit run app.py
2024-10-25 12:15:52 - Created default config file at /workspaces/stocks-project/
2024-10-25 12:15:52 - Created default translation directory at /workspaces/stock
2024-10-25 12:15:52 - Created default translation file at /workspaces/stocks-pro
2024-10-25 12:15:52 - Created default translation file at /workspaces/stocks-pro
2024-10-25 12:15:52 - Created default translation file at /workspaces/stocks-pro
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2024-10-25 12:15:52 - Created default translation file at /workspaces/stocks-pro
2024-10-25 12:15:52 - Created default translation file at /workspaces/stocks-pro
2024-10-25 12:15:55 - Python REPL can execute arbitrary code. Use with caution.
2024-10-25 12:15:55 - Created default chainlit markdown file at /workspaces/stoc
2024-10-25 12:15:55 - Your app is available at http://localhost:8000
```

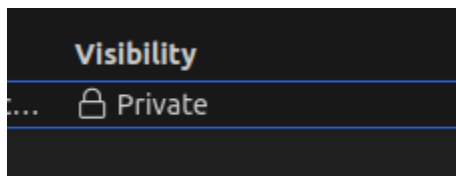
15. App should be accessible at port 8000 of localhost. You have to forward the port to access it in the browser.

Go to the **PORTS** tab beside the **TERMINAL** tab. The port should already be forwarded, else you can click on 'Add Port' and provide the port number.

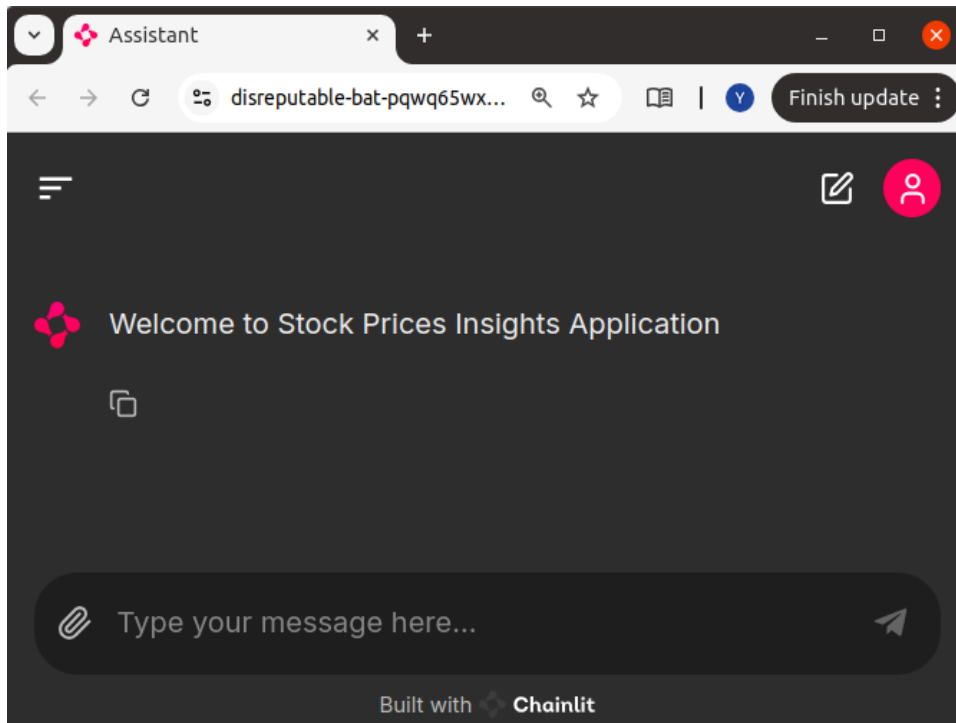


Similarly, you can make the port visibility to 'public' to share it with others.

Do right click on it, then Port Visibility -> Public.



16. Once the port is forwarded, you can access its 'Forwarded Address' to access the api.



17. Try with some user inputs:

Welcome to Stock Prices Insights Application

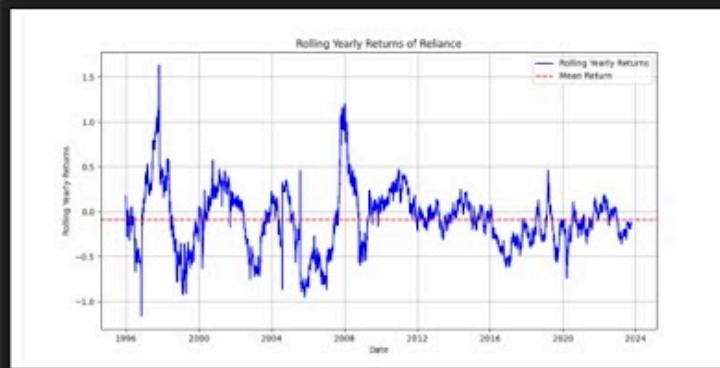


I want to understand the distribution of yearly returns of Reliance.
Can you help me plot the rolling yearly returns highlighting the
mean returns using a mean return line.



Response:

Mean Return: -0.0893



What kind of data do you have



Response:

The database contains financial data for various companies, organized into four main tables.

1. **Stock Prices:** This table records daily stock prices, including the opening, high, low, and closing prices, along with trading volume for each company identified by its symbol.
2. **Income Statement:** This table provides detailed financial performance metrics, such as revenue, expenses, net income, and earnings per share (EPS) for each company, along with relevant dates and currency information.
3. **Balance Sheet Statement:** This table captures a company's financial position at a specific date, detailing assets, liabilities, and equity components, including cash, receivables, and debt.
4. **Cash Flow Statement:** This table outlines the cash inflows and outflows from operating, investing, and financing activities, providing insights into a company's cash management.

Each table is linked by the company symbol, allowing for comprehensive financial analysis.



Further suggestion:

You can ask for insights such as the historical stock price trends for a specific company, the latest income statement details for a particular symbol, or the cash flow performance over the last few quarters. Additionally, you might inquire about the balance sheet metrics to assess a company's financial health.



Type your message here...



Built with  Chainlit

Hello there



Response:

Out of context.



Further suggestion:

You can ask for insights such as:

1. What are the trends in stock prices for a specific company over the last year?
2. How has the net income of a company changed over the past few quarters?
3. What is the current debt-to-equity ratio for a particular company?
4. Can you provide a comparison of revenue and expenses for a company over the last few years?

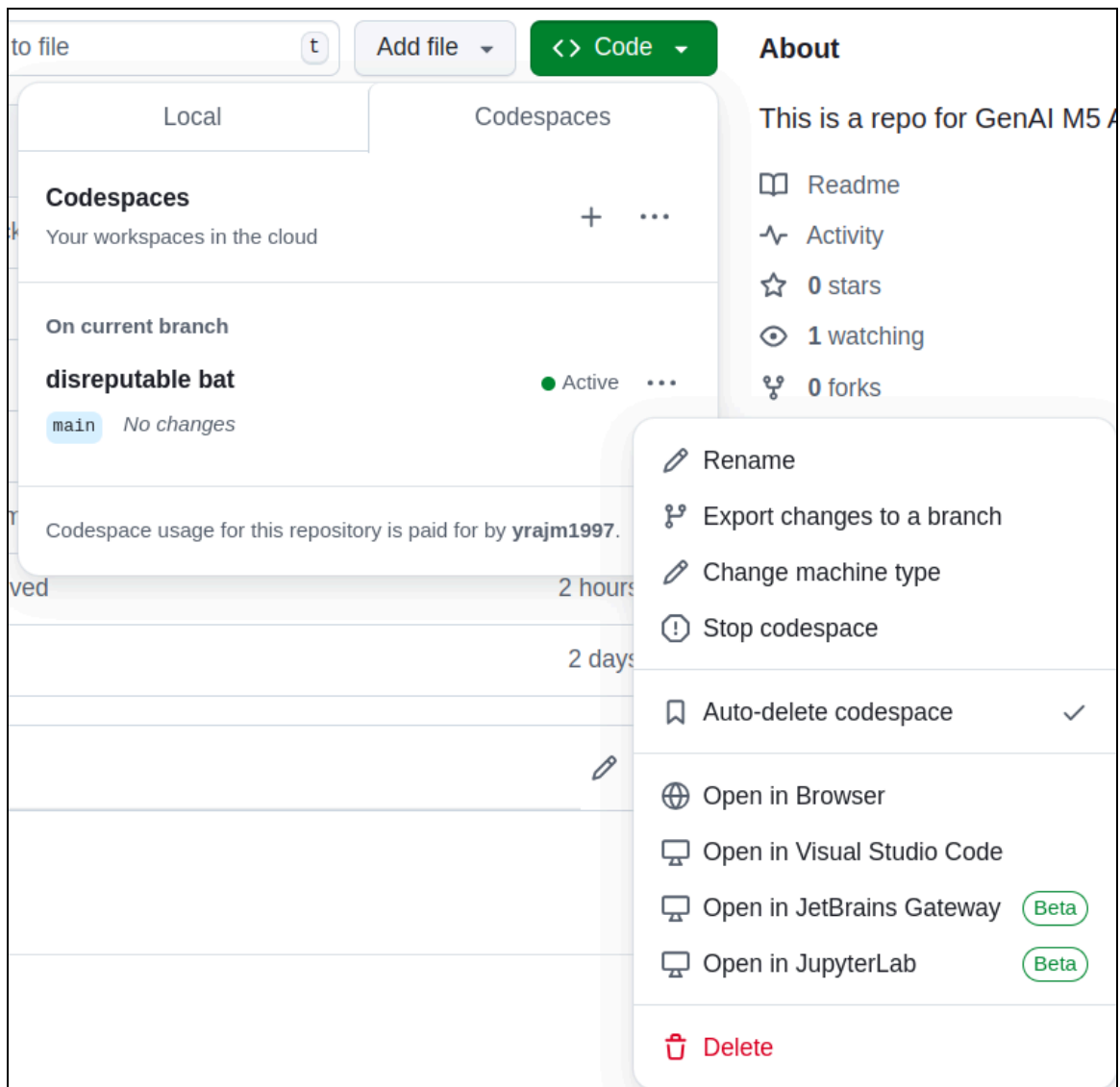


Type your message here...



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18. Once done, close the application by pressing Ctrl + C in the terminal.
19. At last, delete the Codespace.



~~~END~~~